Configurations d’une box

La box est une Globble développée par l'entreprise. Elle possède deux modems, deux ports pour des cartes SIM, un port USB, des ports Ethernet (RJ45) et des ports pour mettre des antennes réseau.

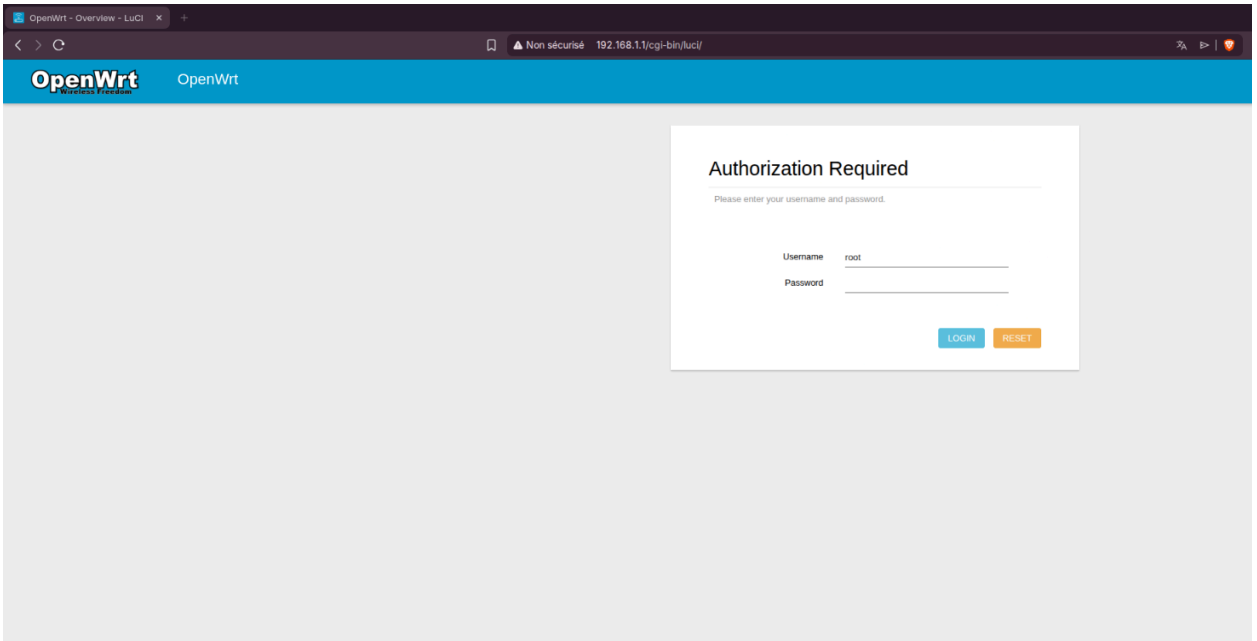
Elle est gérée par un logiciel nommées steer qui est installé dans la box pour pouvoir la configurer et permet également de la gérer à distance

Tout d’abord pour configurer cette box je vérifie que j’y est bien accès depuis mon poste

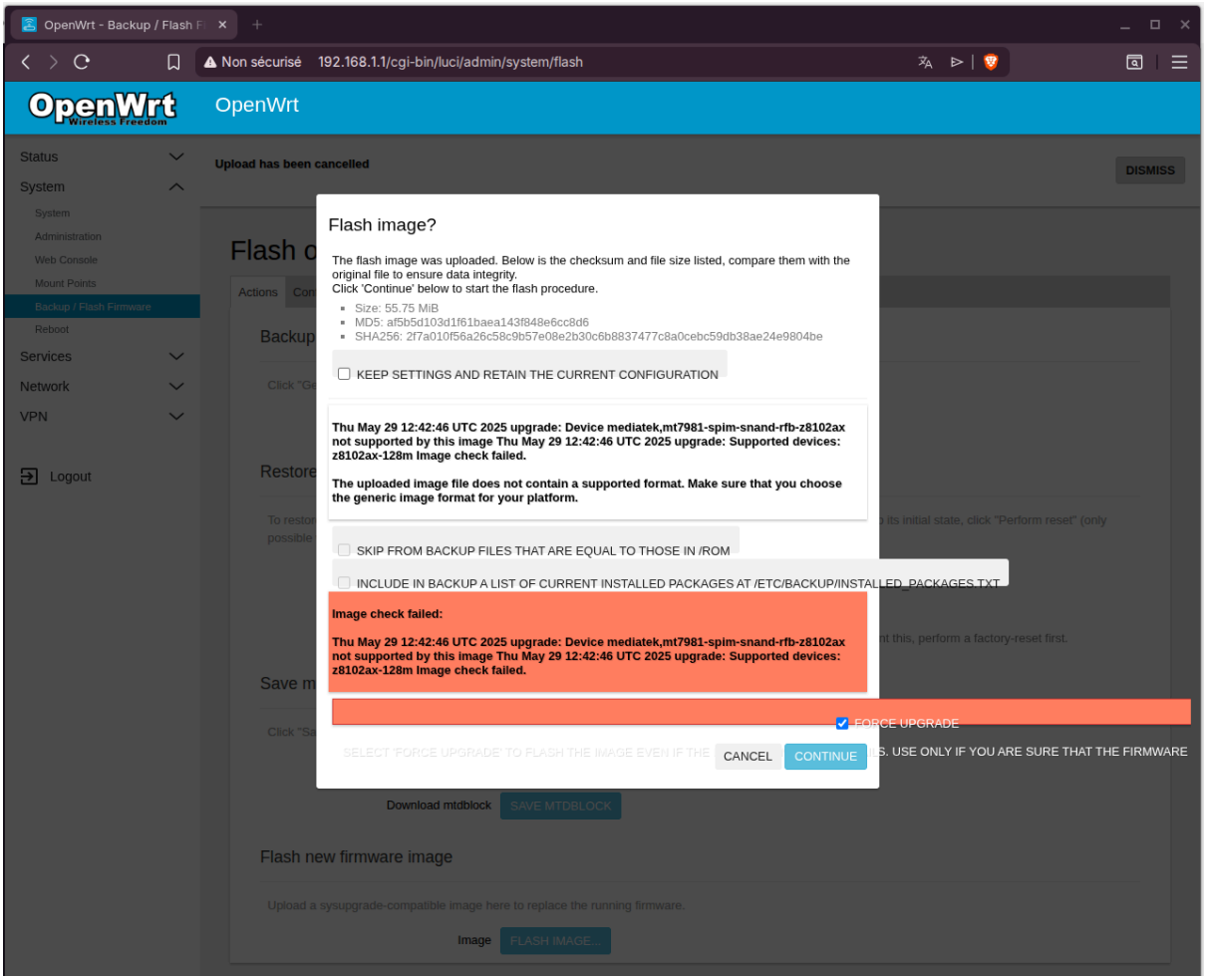
via le terminal

```
melih@Stellar:~$ ssh root@192.168.1.1
The authenticity of host '192.168.1.1 (192.168.1.1)' can't be established.
RSA key fingerprint is SHA256:/h0fVwcOQ7gUSoFDH+Ssu8PcdAfQW36PCKsQ4Sf1CrY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.1' (RSA) to the list of known hosts.
root@192.168.1.1's password:


BusyBox v1.33.2 (2023-06-16 10:11:54 UTC) built-in shell (ash)

  _______                     ________        __
 |       |.-----.-----.-----.|  |  |  |.----.|  |_
 |   -   ||  _  |  -__|     ||  |  |  ||   _||   _|
 |_______||   __|_____|__|__||________||__|  |____|
          |__| W I R E L E S S   F R E E D O M
 -----------------------------------------------------
 OpenWrt 21.02-SNAPSHOT, unknown
 -----------------------------------------------------
root@OpenWrt:~#
```

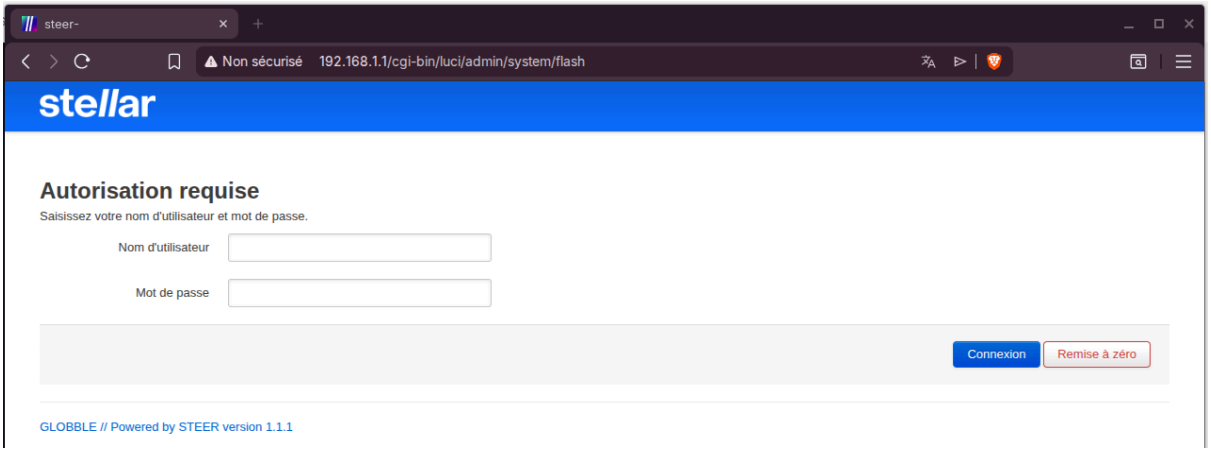
via un moteur de recherche



Ensuite je dois installer le logiciel steer sur la box je upload le fichieret je force son installation et fais supprimer l'ancien



Une fois installer nous avons la nouvelle interface graphique qui s’affiche



et dans le terminal

```
melih@Stellar:~$ ssh root@192.168.1.1
The authenticity of host '192.168.1.1 (192.168.1.1)' can't be established.
ED25519 key fingerprint is SHA256:GpPyBEEnonSAizlHgHG27LaEgxmfrYjWmR/hQntWstY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.1' (ED25519) to the list of known hosts.
root@192.168.1.1's password:


BusyBox v1.37.0 (2025-11-02 15:32:10 UTC) built-in shell (ash)


 OpenWrt recently switched to the "apk" package manager!

 OPKG Command           APK Equivalent      Description
 ---------------------------------------------------------------
 opkg install <pkg>     apk add <pkg>       Install a package
 opkg remove <pkg>      apk del <pkg>       Remove a package
 opkg upgrade           apk upgrade         Upgrade all packages
 opkg files <pkg>       apk info -L <pkg>   List package contents
 opkg list-installed    apk info            List installed packages
 opkg update            apk update          Update package lists
 opkg search <pkg>      apk search <pkg>    Search for packages
 ---------------------------------------------------------------

For more https://openwrt.org/docs/guide-user/additional-software/opkg-to-apk-cheatsheet

root@steer-:~#
```

Donc mtn que l'installation du software sur la box a réussis je connecte 2 cartes sim dessus et je vérifie que les modems fonctionne bien

```
root@steer-f85e3cee141c:~# mmcli -m 0
  ------------------------------------
  General   |                   path: /org/freedesktop/ModemManager1/Modem/0
            |              device id: 5b0e16371fa8929b87703c9900d8151d9946fa46
  ------------------------------------
  Hardware  |           manufacturer: Quectel
            |                  model: RM520N-GL
            |       firmware revision: RM520NGLAAR03A03M4G_BETA_20240522C
            |          carrier config: ROW_Commercial
            | carrier config revision: 0A010809
            |           h/w revision: RM520N-GL
            |              supported: gsm-umts, lte, 5gnr
            |                current: gsm-umts, lte, 5gnr
            |           equipment id: 868371054210337
  ------------------------------------
  System    |                 device: /sys/devices/platform/soc/11200000.usb/usb2/2-1/2-1.1
            |                physdev: /sys/devices/platform/soc/11200000.usb/usb2/2-1/2-1.1
            |                drivers: cdc_mbim, option1
            |                 plugin: quectel
            |           primary port: cdc-wdm0
            |                  ports: cdc-wdm0 (mbim), ttyUSB0 (ignored), ttyUSB1 (gps),
            |                         wwan0 (net)
  ------------------------------------
  Status    |                   lock: sim-pin2
            |         unlock retries: sim-pin2 (3)
            |                  state: connected
            |            power state: on
            |            access tech: lte, 5gnr
            |         signal quality: 18% (recent)
  ------------------------------------
```

```
root@steer-f85e3cee141c:~# mmcli -m 1
  ------------------------------------
  General   |                   path: /org/freedesktop/ModemManager1/Modem/1
            |              device id: 04447658fd843217f40e98bc89c4e2953363d417
  ------------------------------------
  Hardware  |           manufacturer: Quectel
            |                  model: RM520N-GL
            |       firmware revision: RM520NGLAAR03A03M4G_BETA_20240522C
            |          carrier config: ROW_Commercial
            | carrier config revision: 0A010809
            |           h/w revision: RM520N-GL
            |              supported: gsm-umts, lte, 5gnr
            |                current: gsm-umts, lte, 5gnr
            |           equipment id: 868371054194770
  ------------------------------------
  System    |                 device: /sys/devices/platform/soc/11200000.usb/usb2/2-1/2-1.2
            |                physdev: /sys/devices/platform/soc/11200000.usb/usb2/2-1/2-1.2
            |                drivers: cdc_mbim, option1
            |                 plugin: quectel
            |           primary port: cdc-wdm1
            |                  ports: cdc-wdm1 (mbim), ttyUSB4 (ignored), ttyUSB5 (gps),
            |                         wwan1 (net)
  ------------------------------------
  Status    |                   lock: sim-pin2
            |         unlock retries: sim-pin2 (3)
            |                  state: connected
            |            power state: on
            |            access tech: lte, 5gnr
            |         signal quality: 0% (cached)
  ------------------------------------
```

Donc on peut voir que les deux modems fonctionnent et qu'ils sont bien connectés à internet.

Ensuite j’ai connecté une antenne Starlink sur la prise WAN

Je met la WAN en master pour quel soit privilégié et met les deux autre en backup pour qu’il prennent le relais pour finir de la connexion

WAN

Multipath TCP: master
One interface must be set as master

Weight: 100
Only for *weight schedulers/path managers (if any available)
A weight >100 make it more attractive, a weight <100 make it less attractive. Max 256

CELLULAR1

Multipath TCP: backup
One interface must be set as master

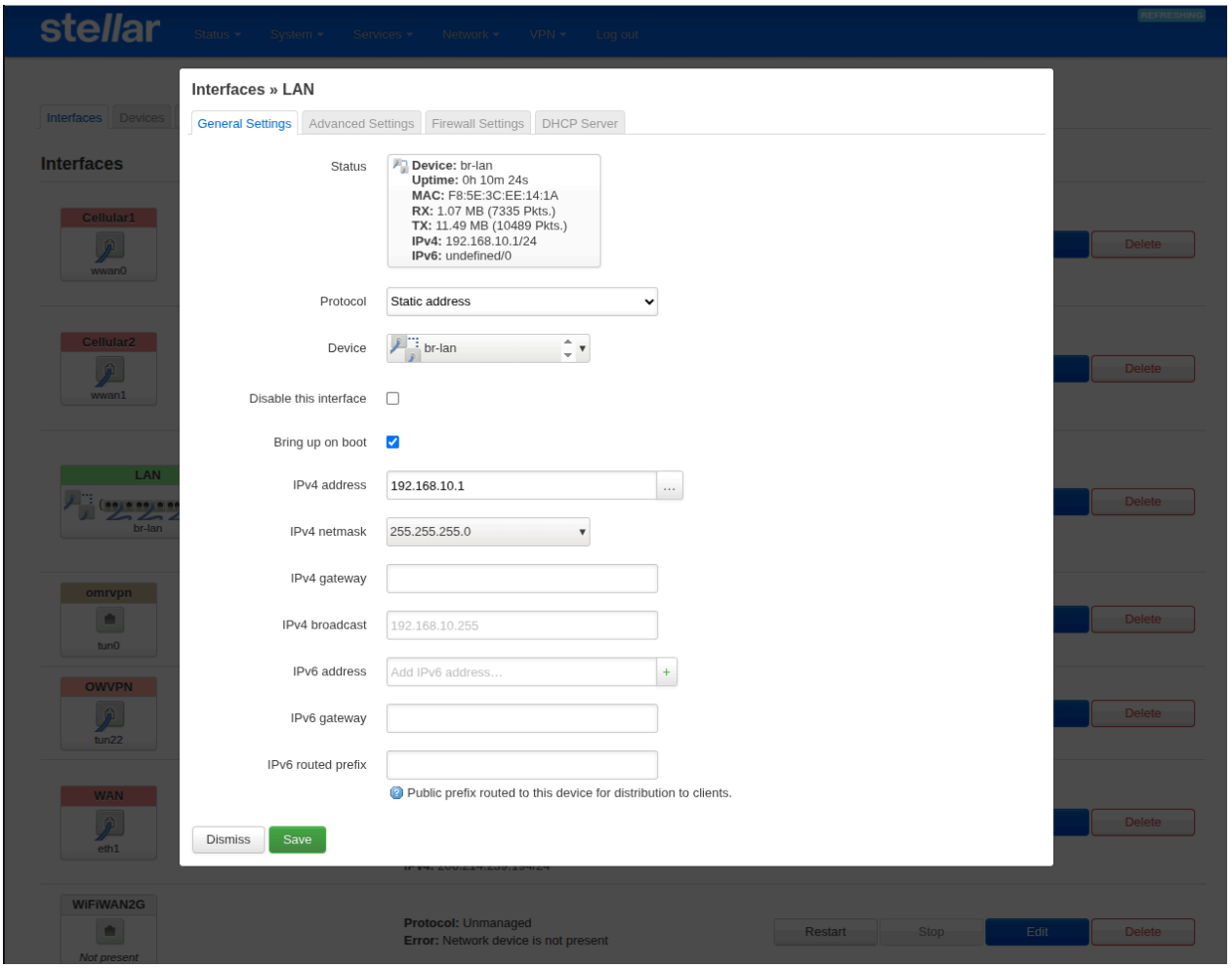
Weight: 100
Only for *weight schedulers/path managers (if any available)
A weight >100 make it more attractive, a weight <100 make it less attractive. Max 256

CELLULAR2

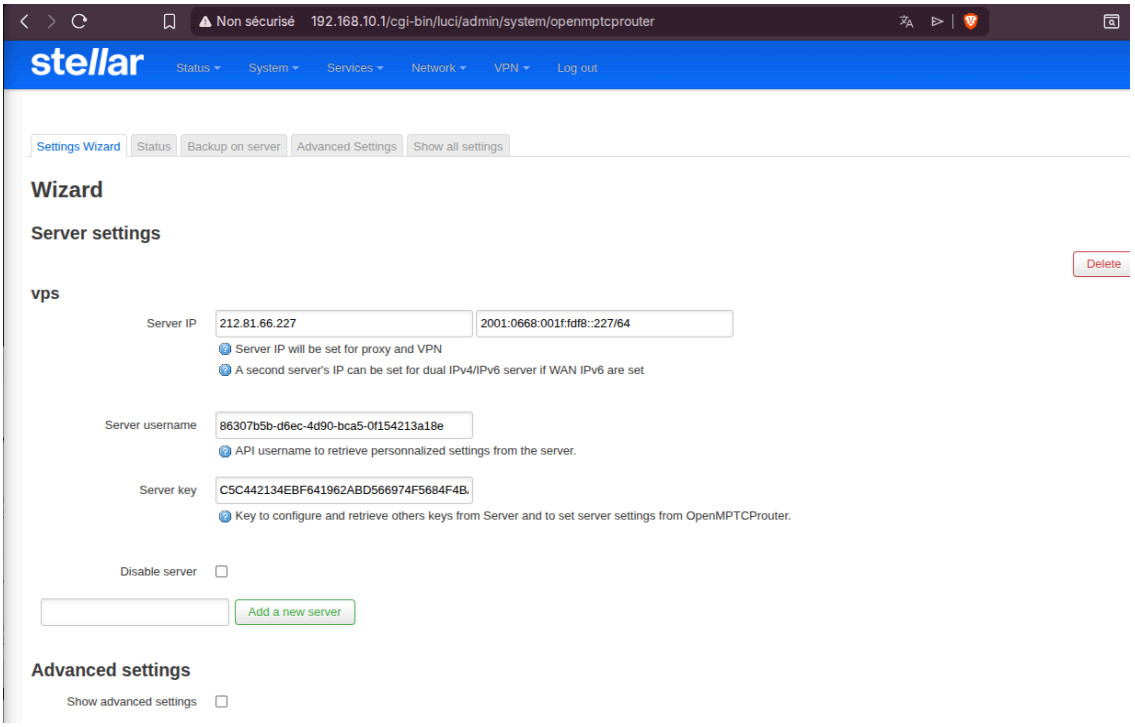
Multipath TCP: backup
One interface must be set as master

Weight: 100
Only for *weight schedulers/path managers (if any available)
A weight >100 make it more attractive, a weight <100 make it less attractive. Max 256

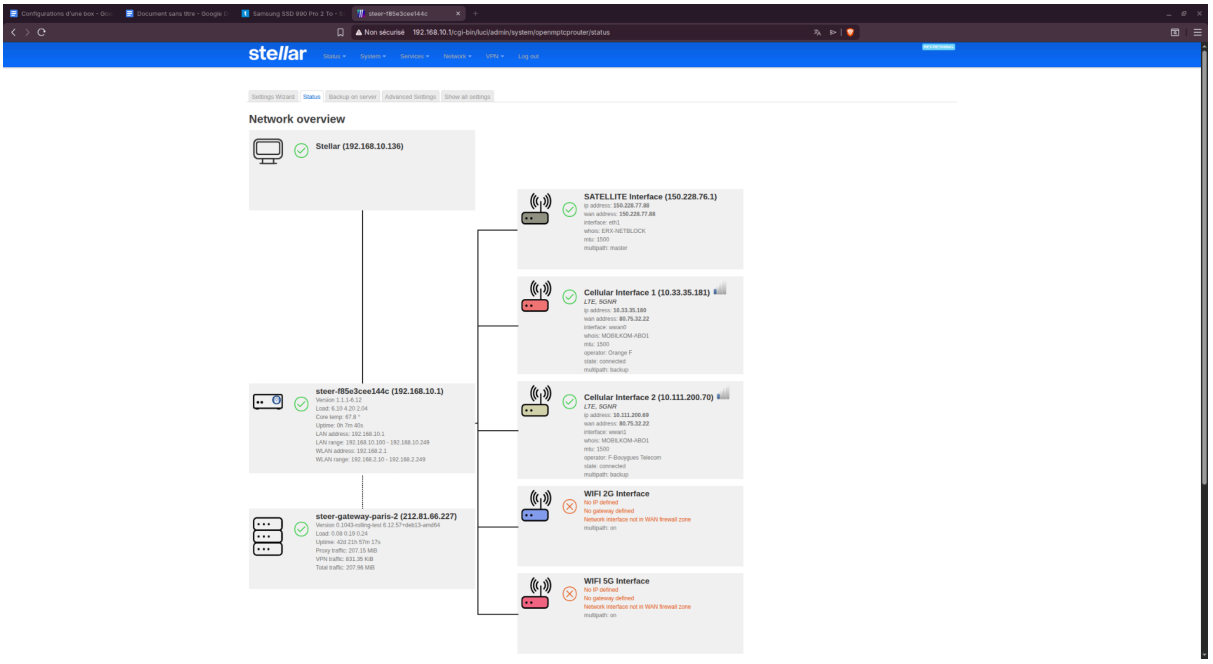
Je change l'ip de la box



puis un collègue ma configurer le vps



On vérifie sur la box que tout marche bien



En essayant aussi des ping un par un chaque carte réseau connectée

```
root@steer-f85e3cee141c:~# ping -I wwan0 radionugget.com
PING radionugget.com (216.198.79.1) from 10.33.116.11 wwan0: 56(84) bytes of data.
64 bytes from 216.198.79.1: icmp_seq=1 ttl=249 time=55.8 ms
64 bytes from 216.198.79.1: icmp_seq=2 ttl=249 time=74.0 ms
64 bytes from 216.198.79.1: icmp_seq=3 ttl=249 time=71.4 ms
^C
--- radionugget.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 55.841/67.075/73.961/8.011 ms
```

```
root@steer-f85e3cee141c:~# ping -I wwan1 radionugget.com
PING radionugget.com (216.198.79.1) from 10.115.241.61 wwan1: 56(84) bytes of data.
64 bytes from 216.198.79.1: icmp_seq=1 ttl=249 time=65.8 ms
64 bytes from 216.198.79.1: icmp_seq=2 ttl=249 time=57.6 ms
64 bytes from 216.198.79.1: icmp_seq=3 ttl=249 time=74.1 ms
^C
--- radionugget.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 57.649/65.854/74.117/6.723 ms
```

```
root@steer-f85e3cee141c:~# ping -I eth1 8.8.8.8
PING 8.8.8.8 (8.8.8.8) from 150.228.77.88 eth1: 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=26.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=17.1 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=21.1 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 3 received, 40% packet loss, time 4024ms
rtt min/avg/max/mdev = 17.105/21.574/26.551/3.873 ms, pipe 2
```

et enfin pour la dernière vérification je me connecte sur mon téléphone a la box pour tester la connexion grâce à un speedtest